

Homework12

August 31, 2026

```
[1]: import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt

titanic = sns.load_dataset("titanic")
titanic.info() #Check the data types of the columns
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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   survived        891 non-null    int64
 1   pclass          891 non-null    int64
 2   sex             891 non-null    object
 3   age            714 non-null    float64
 4   sibsp          891 non-null    int64
 5   parch          891 non-null    int64
 6   fare           891 non-null    float64
 7   embarked       889 non-null    object
 8   class          891 non-null    category
 9   who            891 non-null    object
10  adult_male     891 non-null    bool
11  deck          203 non-null    category
12  embark_town    889 non-null    object
13  alive          891 non-null    object
14  alone          891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

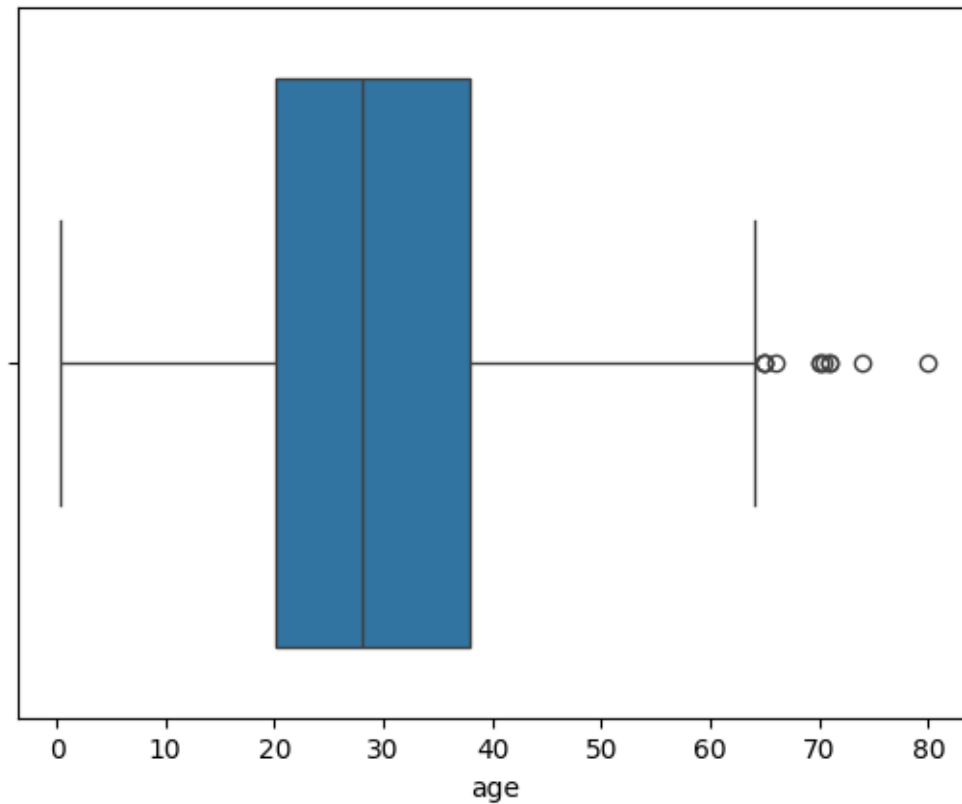
```
[2]: #Box plot for "age" and compute the IQR
sns.boxplot(data=titanic, x="age")
plt.show()

age = titanic["age"].dropna()
Q1, Q2, Q3 = np.percentile(age, [25, 50, 75], method='linear')
IQR = Q3 - Q1
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min = Q1 - 1.5 * IQR
max = Q3 + 1.5 * IQR
Q1, Q2, Q3, IQR, min, max

```



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[2]: (np.float64(20.125),
      np.float64(28.0),
      np.float64(38.0),
      np.float64(17.875),
      np.float64(-6.6875),
      np.float64(64.8125))

```

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[3]: #The outliers for "age"
      titanic[titanic["age"] > max]

```

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[3]:
   survived  pclass  sex  age  sibsp  parch  fare  embarked  class  \
33         0      2  male  66.0      0      0  10.5000         S  Second
54         0      1  male  65.0      0      1  61.9792         C   First
96         0      1  male  71.0      0      0  34.6542         C   First
116        0      3  male  70.5      0      0   7.7500         Q  Third
280        0      3  male  65.0      0      0   7.7500         Q  Third
456        0      1  male  65.0      0      0  26.5500         S   First

```

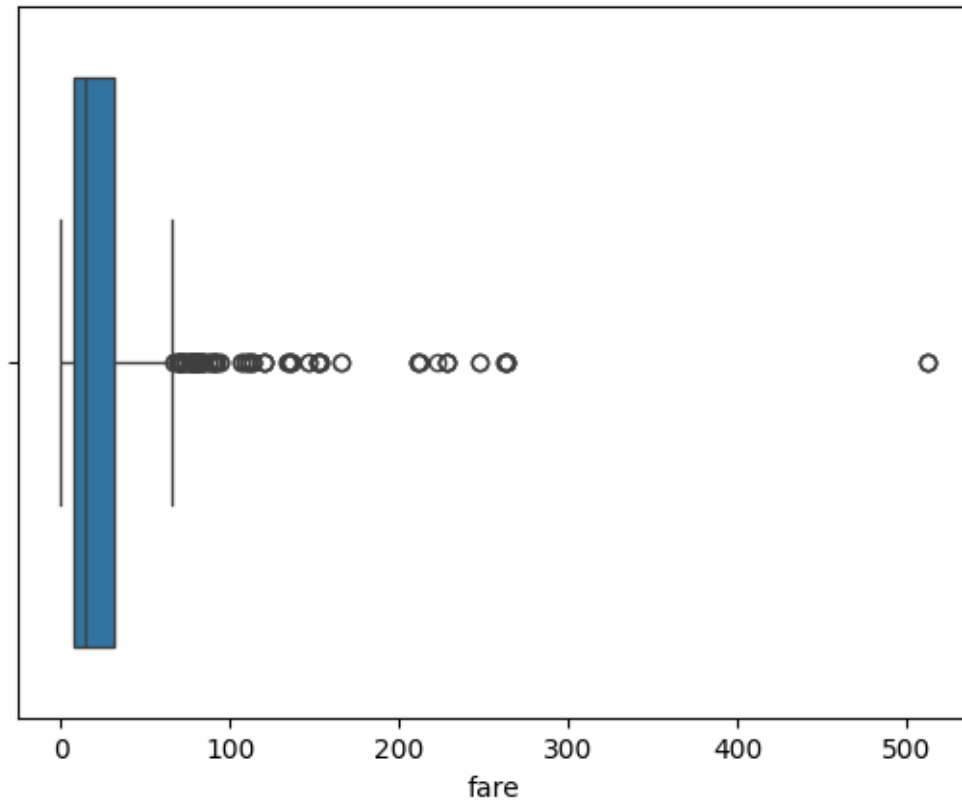
493	0	1	male	71.0	0	0	49.5042	C	First
630	1	1	male	80.0	0	0	30.0000	S	First
672	0	2	male	70.0	0	0	10.5000	S	Second
745	0	1	male	70.0	1	1	71.0000	S	First
851	0	3	male	74.0	0	0	7.7750	S	Third

	who	adult_male	deck	embark_town	alive	alone
33	man	True	NaN	Southampton	no	True
54	man	True	B	Cherbourg	no	False
96	man	True	A	Cherbourg	no	True
116	man	True	NaN	Queenstown	no	True
280	man	True	NaN	Queenstown	no	True
456	man	True	E	Southampton	no	True
493	man	True	NaN	Cherbourg	no	True
630	man	True	A	Southampton	yes	True
672	man	True	NaN	Southampton	no	True
745	man	True	B	Southampton	no	False
851	man	True	NaN	Southampton	no	True

```
[4]: #Box plot for "fare" and compute the IQR
sns.boxplot(data=titanic, x="fare")
plt.show()

fare = titanic["fare"].dropna()
Q1, Q2, Q3 = np.percentile(fare, [25, 50, 75], method='linear')
IQR = Q3 - Q1

min = Q1 - 1.5 * IQR
max = Q3 + 1.5 * IQR
Q1, Q2, Q3, IQR, min, max
```



```
[4]: (np.float64(7.9104),
      np.float64(14.4542),
      np.float64(31.0),
      np.float64(23.0896),
      np.float64(-26.724),
      np.float64(65.6344))
```

```
[5]: #The outliers for "fare"
      titanic[titanic["fare"] > max]
```

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[5]:
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	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
1	1	1	female	38.0	1	0	71.2833	C	First	
27	0	1	male	19.0	3	2	263.0000	S	First	
31	1	1	female	NaN	1	0	146.5208	C	First	
34	0	1	male	28.0	1	0	82.1708	C	First	
52	1	1	female	49.0	1	0	76.7292	C	First	
..	...	...	...	...	...	...	...	...	...	
846	0	3	male	NaN	8	2	69.5500	S	Third	
849	1	1	female	NaN	1	0	89.1042	C	First	
856	1	1	female	45.0	1	1	164.8667	S	First	
863	0	3	female	NaN	8	2	69.5500	S	Third	

```
879          1          1 female 56.0          0          1 83.1583          C First
```

	who	adult_male	deck	embark_town	alive	alone
1	woman	False	C	Cherbourg	yes	False
27	man	True	C	Southampton	no	False
31	woman	False	B	Cherbourg	yes	False
34	man	True	NaN	Cherbourg	no	False
52	woman	False	D	Cherbourg	yes	False
..	...	...	...	...	...	...
846	man	True	NaN	Southampton	no	False
849	woman	False	C	Cherbourg	yes	False
856	woman	False	NaN	Southampton	yes	False
863	woman	False	NaN	Southampton	no	False
879	woman	False	C	Cherbourg	yes	False

```
[116 rows x 15 columns]
```

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[ ]:
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```
[6]: #The outliers for "age"
outliers_age = titanic[titanic["age"] > max]
len(outliers_age)
```

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[6]: 8
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[7]: #The outliers for "fare"
outliers_fare = titanic[titanic["fare"] > max]
len(outliers_fare)
```

```
[7]: 116
```